Teacher Guide

Project Title: Ama's Design Challenge: An Interactive Multimedia Learning Package on Elements, Principles, and Colour Theory for SHS Visual Arts

Target Audience: Senior High School Visual Arts Students (Year 1, Ghana)

Duration: 2–3 double periods (80–120 minutes)

Platform: Offline-friendly website (HTML/CSS/JS)

1. Introduction

This Teacher Guide supports the use of an interactive multimedia package designed to teach Elements of Design, Principles of Design, and Colour Theory in the SHS Visual Arts curriculum. The package integrates Ama's Biscuit Packaging Challenge, a competency-based scenario where learners apply theory to solve a real-world design problem.

The package is learner-centered, combining visual lessons, interactive quizzes, and reflection tasks. It encourages creativity, critical thinking, collaboration, and digital literacy as outlined in the Senior High School Curriculum.

2. Learning Objectives

By the end of the lessons and activities, learners will be able to:

1. Identify and apply elements of design (line, shape, texture, value, space).
2. Demonstrate understanding of principles of design (balance, emphasis, harmony, rhythm).
3. Differentiate and use colour schemes (analogous, complementary, triadic) to express mood and meaning.
4. Critically analyze rejected packaging designs and propose solutions.
5. Collaborate in groups to make design decisions.
6. Reflect on their creative choices and articulate design reasoning.

3. Preparation for Teachers

1. Ensure the multimedia package (HTML/CSS/JS files) is copied to the classroom laptop or lab computers.
2. Check that all links (index.html → learn.html → quizzes → Ama's Challenge) work offline.
3. Arrange learners in small groups (3–5) for collaborative tasks.

4. Provide stationery (sketch pads, pencils, markers) for optional offline extension activities.

4. Suggested Lesson Sequence

Step 1: Introduction (10 minutes)

Introduce Ama's Challenge: a young SHS student designing biscuit packaging. Present the problem: three rejected designs with issues in colour, balance, emphasis, and harmony.

Step 2: Learning Modules (30–40 minutes)

Guide learners through:

1. Elements of Design lessons: Element-Intro, Shape, Texture, Value, White Space.
2. Principles of Design lessons: Principles Intro, Balance, Emphasis, Harmony, Rhythm.
3. Colour Theory lessons: Color theory Intro, Wheel, Mood, Analogous, Triadic, Contemporary color usage.
4. Each lesson includes interactive checkpoints to check understanding and offline exercise to be completed by the students, guided by the teacher.

Step 3: Quizzes (30 minutes)

Learners take:

1. Combined Elements & Principles Quiz (elements-quiz.html, which combines elements and principles quizes).
2. Colour Theory Quiz (color-theory-quiz.html).
3. Scores appear on the learner dashboard (dashboard_new.html).

Step 4: Ama's Challenge (30 minutes)

Learners watch and or read Ama's packaging story video and text (ama.mp4 in ama-challenge.html). They complete tasks:

Select all the design sample and give their suggested mistakes and propose ways to solve them and sumit to their teacher and download and print to the teacher or the teacher can access it from the teacher dashboard.

The student are required to complete an offline assignment which would be assessed by the teacher and theclick on done to show he has completed the Challenge.

Step 5: Group Collaboration (10–15 minutes)

In group.html, learners form study groups to compare reflections and propose improvements. Teacher facilitates discussion linking back to curriculum goals.

5. Assessment Guidelines

Formative Assessment:

1. Mini checkpoint quizzes and offline exercises in each lesson.
2. Class discussions and group reflections.

Summative Assessment:

1. Elements & Principles Quiz.
2. Colour Theory Quiz.
3. Ama's Challenge performance (interactive + written reflection).

Sample Rubric:
Knowledge of Elements/Principles: Excellent (4) → Weak (1)

Application of Colour Theory: Excellent (4) → Weak (1)

Critical Thinking: Excellent (4) → Weak (1)

Creativity: Excellent (4) → Weak (1)

6. Teacher Tips & Differentiation

1. **For struggling learners:** Pair with stronger peers; provide guiding questions.
2. **For advanced learners:** Challenge them to redesign the biscuit package using Canva (ICT tool and digital competency)or paper.
3. **Inclusivity**: Encourage multiple forms of response (verbal, written, sketching).
4. **Offline adaptation:** If no computers(personal laptops, mobile device like phone and tablets), project lessons and let learners vote on answers.

7. Challenges & Solutions

1. **Limited devices:** Use projector and group learners.
2. **Technical issues:** Keep paper-based backup quizzes.
3. **Time constraints:** Prioritize core lessons and Ama's Challenge; assign reflection and at-the-end lesson exercises as homework.

8. Conclusion

This package enables SHS learners to move from theory to practice by connecting abstract design concepts to a real-world problem (eg. Biscuit package). It strengthens their ability to analyze, critique, and create, fulfilling the competency-based curriculum objectives. Teachers are encouraged to adapt the sequence and activities to suit their classroom context.

Student Dashboard (dashboard_new.html)

Purpose: Helps learners **track their own learning progress**, view quiz scores, and monitor group participation.

Visual Layout:

- **Header/Navigation:** Shows the Visual Arts logo, links to *Home, Learn, Solve*, and a dropdown with *Profile, Settings, Logout*.
- **Welcome Message:** Personalized with learner's name ("Welcome, Ama").

Sections & Functions:

1. **Progress Summary**
    - **Lessons Completed:** Out of 3 (Elements & Principles, Colour Theory, Ama's Challenge).
    - **Quiz Scores:** Average percentage across all quizzes.
    - **Group:** Displays current group name or "None."
2. **Continue Learning/Start Learning**
    - Displays 3 "lesson cards":
        - *Balance & Harmony*
        - *Emphasis in Design*
        - *Colour Theory & Mood*
    - Each card shows:
        - Lesson description.
        - Progress bar (0%, 50%, or 100%).
        - "Continue" button linking directly to the relevant section in *learn.html*.
3. **Recent Activity**
    - Logs completed lessons, quizzes, and scores (e.g., "Completed Colour Theory Quiz – 80%").
    - If no activity, shows "No recent activity yet."

Functionality:

- Automatically updates lesson progress after completing checkpoints.
- Calculates average quiz score from stored attempts.
- Displays **group membership** if the student joined via group.html.
- Stores user info (name, group code, quiz scores, progress) in **localStorage**.
- Logout clears stored data and redirects to login.html.

Teacher Dashboard (teacher-dashboard.html)

Purpose: Provides teachers with a **class-level overview** of student engagement, performance, and collaboration.

Visual Layout:

- **Header/Navigation:** Visual Arts logo + links similar to student view, but role-specific (teacher tools).
- **Welcome Message:** "Welcome, Teacher [Name]."

Sections & Functions:

1. **Class Progress Overview**
    - Aggregates student data (from localStorage or class list).
    - Displays statistics:
        - Number of students who completed each lesson.
        - Average scores for Elements & Principles Quiz, Colour Theory Quiz.
        - Percentage of learners who attempted Ama's Challenge.
2. **Group Collaboration Tracker**
    - Lists group codes and group members.
    - Shows group activity logs (reflections submitted, group quiz attempts).
    - Helps teacher identify active vs inactive groups.
3. **Recent Class Activity**
    - Timeline-style log (e.g., "Ama completed Balance Lesson – 100%," "Kwame scored 85% in Colour Theory Quiz").
    - Teachers can click entries to see student details.
4. **Teacher Tools**
    - Export class performance summary (e.g., to CSV or PDF).
    - Reset class progress if needed for a new session.

Functionality:

- Pulls aggregated learner data from **localStorage** (in practice, could be synced to a database).
- Provides **snapshot analytics**: lesson completion rates, quiz averages, participation.
- Supports **formative assessment** by showing where most students struggled.
- Allows teacher to **monitor group collaboration**, not just individual scores.

Summary:

- The **Student Dashboard** is **personal**: it shows *their own* progress, scores, and activity log.
- The **Teacher Dashboard** is **aggregated**: it shows *the class as a whole*, group participation, averages, and recent activities.